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MILITARY SPECIFICATION

# CARTRIDGE, IGNITION, M285
# LOADING, ASSEMBLING AND PACKING

## 1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of ignition cartridge designated as Cartridge, Ignition, M285.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| UU-T-106 | -Tape, Masking; Paper (Pressure Sensitive) |
| PPP-B-566 | -Boxes, Folding Paperboard |
| PPP-B-636 | -Box, Fiberboard |
| PPP-B-676 | -Boxes, Set-Up, Paperboard |

MILITARY

| | |
|---|---|
| MIL-P-116 | -Preservation, Methods of |
| MIL-B-117 | -Bags, Interior Packaging |
| MIL-B-2427 | -Boxes, Ammunition Packing, Wood, Nailed |
| MIL-A-2550 | -Ammunition and Special Weapons, General Specification for |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | -Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105) |
| MIL-STD-109 | -Quality Assurance Terms and Definitions |

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MIL-STD-414 -Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1168 -Lot Numbering of Ammunition
MIL-STD-1235 -Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMY

8796522 -Marking Diagram and Sealing for Wood Packing Boxes
9240960 -Cartridge, Ignition

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-90 -Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Order for the above publications should cite "49 CFR 71-90 (latest revisions)".)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-The materials and parts shall be in accordance with applicable specifications and drawings.

3.2 Assembly.-The cartridge assembly shall comply with all requirements specified on Drawing (dwg.) 9240960 and with all requirements in applicable specifications.

3.3. Bursting strength.-The cartridge assemblies shall not burst at a compression load of 53 pounds, minimum, but shall burst at a compression load of 102 pounds, maximum, when tested as specified in 4.4.2.



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3.4 Velocity and functioning (Proving ground).-The cartridge assemblies shall function and shall impart a corrected muzzle velocity of not more than 220 feet per second (fps) and not less than 200 fps when tested as specified in 4.4.4. In addition, the standard deviation of the lot shall be not greater than 2.4 fps.

3.5 Tape Wrapping.-The tape shall enclose the full length of the cartridge outside diameter. The minimum tape overlap specified on drawing 9240960 shall be maintained for the full length of the cartridge. Tape shall extend over the top of the roll crimp for a full 360 degrees. The tape shall contain no cuts or missing pieces. There shall be no creases or blisters in the tape on the outside diameter of the cartridge nor shall the tape overlap the end of the cartridge to the extent that any portion of the flash hole is covered.

3.6 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, corrosion and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.7 The propellant shall be assessed to achieve a velocity of 210 fps.

3.8 First article inspection.-This specification makes provisons for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purcahse order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to



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the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Initial production inspection .-The contractor, prior to initiation of regular production, shall submit an initial production sample consisting of loaded assemblies, inert parts and sub-assemblies in the quantities specified herein, which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis and compliance. An initial production sample, or portion thereof, as directed by the contracting officer, shall also be submitted, whenever there is a lapse in excess of 90 days; or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply such as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer.

4.2.1 Inert parts

4.2.1.1 Submission-Twenty-five (25) complete sets of inert parts (i.e. 25 each of every inert component and inert



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sub-assembly) shall be submitted for inspection to a Government approved laboratory designated by the contracting officer.

4.2.1.2 Inspection to be performed.-The assemblies and components shall be inspected for conformance with any or all of the requirements of this specification and any or all of the requirements of the applicable drawings.

4.2.1.3 Rejection.-If any inert assembly or component fails to comply with any or all of the applicable requirements, the inert parts initial production sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any inert assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit an additional sample or portion thereof of inert assemblies and components for initial production inspection.

4.2.2 Loaded assemblies

4.2.2.1 Submission

4.2.2.1.1 Ignition cartridge assemblies-Fifty (50) ignition cartridge assemblies shall be submitted for testing to a Government proving ground designated by the contracting officer. In addition, one hundred (100) assemblies shall be submitted for examination and tests as Picatinny Arsenal, Dover, New Jersey. All tests shall be performed with the equipment, and in accordance with the methods, specified in 4.4.

4.2.2.2 Inspections to be performed.-The assemblies and components shall be inspected for conformance with any or all of the requirements of this and other applicable specifications and any or all of the requirements of the applicable drawings.

4.2.2.3 Rejection.-If any loaded assembly fails to comply with any or all of the requirements specified, the initial production sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an item in the sample to comply with any or all of the applicable requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit an additional sample or portion thereof of loaded items for initial production inspection. Until that phase of the initial production sample consisting of loaded items is acceptable, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Quality Conformance Inspection



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4.3.1 Lot Formation.-The term "inspection lot", as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawing, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in Standard MIL-STD-105. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects (and Major Defects, when so specified) shall be 100 percent. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of Major and Minor Defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each Major Defect and an AQL of 0.65 percent for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor Defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Body assembly (see dwg. 9243225 covering a detail of dwg. 9240960).



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| Categories | Defects | Method Of Inspection |
|---|---|---|
| Critical: | | |
| 1. | Washer flash hole missing | Visual |
| 2. | Tube and washer assembly in incorrect end according to color of outer tube | Visual |
| Major: | AQL 0.40 percent | |
| 101. | Any component missing, loose or damaged | Visual-Manual |
| 102. | Any component incompletely sealed | Visual-Manual |
| Minor: | AQL 0.40 percent | |
| 201. | Evidence of poor workmanship (see 3.6) | Visual |

4.3.2.2 Cartridge assembly, prior to crimping (see dwg. 9240960).

| Categories | Defects | Method Of Inspection |
|---|---|---|
| Critical: | | |
| 1. | Component punctured or loose allowing charge to escape | Visual |
| Minor: | AQL 0.65 percent | |
| 201. | Foreign matter in charge | Visual |

4.3.2.3 Cartridge assembly (see dwg. 9240960).

| Categories | Defects | Method Of Inspection |
|---|---|---|
| Critical: | | |
| 1. | Fluorescent paper on yellow end | Visual |
| Major: | AQL 0.65 percent | |
| 101. | Length, max | Gage |
| 102. | Diameter, max | Gage |
| 103. | Pressure sensitive tape missing or incorrectly applied (see 3.5) | Visual |
| 104. | Fluorescent paper missing from green end or inverted | Visual |
| Minor: | AQL 2.5 percent | |
| 201. | Washer slanted | Visual |
| 202. | Onion Skin Disc (or Fluorescent type disc as applicable) slanted | Visual |
| 203. | Crimp incomplete or inadequate | Visual |



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204. Outer tube or crimp not coated.....Visual
205. Outer tube color incorrect.....Visual
206. Marking misleading or unidentifiable....Visual
207. Evidence of poor workmanship (See 3.6)..Visual

4.3.2.4 Carton, Prior to sealing (see 5.1.1.1 and 5.1.1.2).

| Categories | Defects | Method Of Inspection |
|---|---|---|
| Critical: | None defined | |
| Major: | AQL 0.40 percent | |
| 101. | Any packing material missing..... | Visual |
| 102. | Incorrect number of cartridges..... | Visual |
| Minor: | None defined | |

4.3.2.5 Carton, sealed (see 5.1.1.1 and 5.1.1.2).

| Categories | Defects | Method Of Inspection |
|---|---|---|
| Critical: | None defined | |
| Major: | None defined | |
| Minor: | AQL 1.5 percent | |
| 201. | Sealing strip torn, badly wrinkled or incomplete..... | Visual |
| 202. | Carton torn, cut or punctured..... | Visual |
| 203. | Contents move excessively upon shaking.. | Manual |
| 204. | Marking misleading or unidentifiable (see 5.3)..... | Visual |

4.3.2.6 Sealed bag (see 5.1.1.1).

| Categories | Defects | Method Of Inspection |
|---|---|---|
| Critical: | None defined | |
| Major: | AQL 0.40 percent | |
| 101. | Seal improper or incomplete..... | Visual |
| 102. | Bag torn or otherwise mutilated..... | Visual |
| Minor: | AQL 1.0 percent | |
| 201. | Seal improperly located..... | Visual |
| 202. | Marking misleading or unidentifiable (see 5.3)..... | Visual |



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4.3.2.7 Packing (see 5.2.1)

| Categories | Defects | Method Of Inspection |
|---|---|---|
| Critical: | None defined | |
| Major: | AQL 0.65 percent | |
| 101. | Box broken, board loose, or protruding nail | Visual |
| 102. | Hardware or strapping missing, broken or loose | Visual-Manual |
| 103. | Carton missing | Visual |
| Minor: | AQL 1.5 percent | |
| 201. | Car seal missing, unsealed or improperly positioned | Visual |
| 202. | Hardware or strapping improperly engaged or mislocated | Visual |
| 203. | Marking misleading, missing or unidentifiable | Visual |
| 204. | Contents loose | Visual-Manual |

4.3.3 Testing

4.3.3.1 Bursting strength (see 3.3) Major defect-SEE FIGURE 1.

4.3.3.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, the sampling plan and procedure for this test shall be in accordance with MIL-STD-105 Code Letter J and AQL of 0.65 percent. The test shall be performed as specified in 4.4.2.

4.3.3.1.2 After three consecutive lots have met the criteria of 4.3.3.1.1, the sampling plan for this test shall be in accordance with MIL-STD-105 Code Letter F, using an AQL of 0.65 percent. The test shall be performed as specified in 4.4.2.

4.3.3.2 Weight of Propellant (see dwg. 9240960) weight above max, Major defect. Weight below minimum (min) Critical defect.-The propellant charge shall be weighed or volumetrically dispensed into a tared cup and then check-weighed 100 percent. The initial weighing or dispensing and check-weighing shall be performed by different operators. Weight limits for check weighing should be established within the applicable weight requirement, depending on the accuracy and precision of the weighing system, to assure compliance with charge weight requirements. Any charge failing to meet the applicable requirement



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shall be removed from the line and returned to a bulk propellant container. Extreme caution shall be exercised in handling propellant charges after check-weighing to prevent spillage. CD

4.3.3.3 Weight of propellant in cartridge assemblies (see dwg. 9240960) Weight above max, Major defect. Weight below min, Critical defect. 315 sample cartridge assemblies shall be selected from each lot for this test. If four or more samples exceed the maximum charge weight or if any sample is below the minimum charge weight, the lot shall be rejected. Testing shall be as specified in 4.4.1.

4.3.3.4 Adhesive or foreign matter in flash tube (see dwg. 9235613 covering a detail of dwg. 9240960 (see Figure 2) Major defect. For each lot of tube assemblies a sample shall be selected according to MIL-STD-105 using an AQL of 0.40 percent. This test shall be performed as specified in 4.4.3.

4.3.3.5 Velocity and functioning (see 3.4) Major defect-

4.3.3.5.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, the sampling plan and procedure for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code Letter M and an AQL of 1.0 percent. In addition, if the standard deviation of the sample multiplied by the factor 0.81 exceeds the lot requirement, the lot shall be rejected. If any round fails to function for reasons attributable to ignition cartridge failure, excluding primer failures, the lot shall be rejected. This test shall be performed as specified in 4.4.4.

4.3.3.5.2 After three consecutive lots have met the criteria of 4.3.3.5.1, the sampling plan for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code Letter I, and an AQL of 1.0 percent. In addition, if the standard deviation of the sample multiplied by the factor 0.75 exceeds the applicable lot requirement, the lot shall be rejected. If any round fails to function for reasons attributable to ignition cartridge failure, excluding primer failures, the lot shall be rejected. This test shall be performed as specified in 4.4.4.

4.3.3.6 Heat seal test of sealed bag (see 5.1.1.1) Major defect.-Sampling, acceptance criteria and test procedure for this test shall be as prescribed in MIL-P-116 for the "Heat Seal Test". CD

4.3.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed in this



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section is identified under the "Method of Inspection" headings in 4.3.2 or in the appropriate testing paragraph under 4.3.3 (See 6.2).

4.4 Test methods and procedures

4.4.1 Weight of Propellant.-The sample cartridge assemblies shall be cut open and all propellant removed and placed on the pan of a precision or analytical balance. The weight of propellant from each sample shall be determined to the nearest 0.01 grain.

4.4.2 Bursting strength.-The body assembly shall be cut in half lengthwise, the powder charge disposed of, and one of the halves placed in the fixture (see figure 1). Set stop on fixture so that punch is at center line of the body assembly half, at a point 1/2 inch from end of body. Lock stop with set screw. Place fixture in universal tester, punch should be checked to see that it slides freely.

TEST NUMBER 1: NO BURST

Apply 53 pound compression load. Remove body assembly. Item is rejected if evidence of bursting is observed. Testing or puncturing of material shall be considered evidence of bursting.

TEST NUMBER 2: BURSTING

If item is not rejected, replace it in fixture, using opposite end of body and apply 102 pounds compression load. If item shows no evidence of bursting it shall be rejected.

4.4.3 Adhesive or foreign matter in flash tube.-Inspection for adhesive or foreign matter in the tube shall be accomplished by pulling tube from disc, sectioning tube longitudinally and examining visually for presence of adhesive or foreign matter on the interior wall of the flash tube. A 360° fillet of glue inside either end of the flash tube will be permitted provided the fillet does not extend inward from the inner surface of the tube in excess of 0.015 inch. Locations and amount of glue permitted are indicated by Figure 2. Glue is not allowable in any inside area of the flash tube except as indicated in Figure 2.



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4.4.4 Velocity and functioning.-The sample M285 ignition cartridge assemblies shall be assembled with M170 fin assemblies and M71A2 primers to 81MM M374 Shell with spiral obturating bands (heat weld not necessary) and inert loaded to 9.12 pounds plus or minus .02 pounds including weight of dummy fuze. The complete assembly less propellant increments shall be fired from a mortar for which the round is standard at an elevation of 45 degrees. Percussion primers used in the sample cartridge shall be not more than two years old. Reference rounds shall be fired using reference standard components and a calibration lot of ignition cartridges for velocity correction.

4.4.5 Propellant charge weight assessment.-Each lot of propellant shall be assessed. Ten assessment cartridges shall be loaded at charge weights of 103, 108 and 113 grains respectively, with a tolerance of plus or minus 0.5 grains. Cartridges shall be loaded using ignition cartridge paper components manufactured by the production supplier. The assessment cartridges shall be assembled to complete rounds made up of reference components inert loaded to 9.12 pounds plus or minus .02 pounds. The rounds shall be conditioned at 70°F for 24 hours prior to firing and shall be fired from a standard 81MM mortar. Reference rounds shall be fired alternately with the three assessment loads for velocity correction. The mean corrected velocities of the three groups shall be used to establish the charge weight velocity relationship and the charge weight assessment shall be determined to the nearest grain. A charge assessment in excess of 115 grains shall not be used without prior approval of Picatinny Arsenal.

## 5. PREPARATION FOR DELIVERY

### 5.1 Packaging

#### 5.1.1 Level A

5.1.1.1 Unit packaging.-Twenty-five (25) cartridges shall be packaged on end in a folding or set-up paperboard box with half-slotted partitions assembled to form twenty-five individual cells in a 5 by 5 arrangement. The box shall conform to PPP-B-566, Variety 1, Style II, or PPP-B-676, Style I, Variety 1. Partitions shall be constructed of minimum .030 thick commercial chipboard or molded of high density polythylene. Cell size shall be 13/16 by 13/16 by 2-1/8 inches in depth inside dimensions. Covers shall be secured to the box by one or more pieces of tape, UU-T-106, 1 inch wide applied to the side and bottom of the closed box. Each box shall be placed in a polyethylene-coated kraft bag conforming to MIL-B-117, Class C. The bag shall be closed by heat sealing.



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5.1.1.2 Intermediate packaging.-Twelve (12) unit packages shall be packed in a corrugated carton conforming to Specification PPP-B-636 W6C or W5C. Carton closure shall be in accordance with the Appendix of the box specification.

5.1.2 Level C.-Unless otherwise specified, the cartridge assemblies shall be delivered in standard commercial containers so constructed as to insure acceptance by common or other carrier for safe transportation, at the lowest rate to the point of delivery. The packing containers shall comply with the requirements of Code of Federal Regulations 49 CFR 71-90 and shall be sufficiently rigid to withstand transportation and permit storing in tiers of at least 10 feet high without damage to container or contents.

5.2 Packing

5.2.1 Level A.-Four intermediate cartons, 1200 cartridges, shall be packed in a packing box constructed in accordance with Specification MIL-B-2427 for a Grade A, Type II, Class 2 box. Boxes shall be closed and strapped in accordance with the box specification.

5.2.2 Level C.-Packing shall be in accordance with the requirement specified in 5.1.2.

5.3 Marking-Each unit box shall be labeled, stamped or printed and each bag printed or stamped with the following information:

25 - Cartridge Ignition, M285
Lot No.
Date Loaded - Month/Year

Letters and numbers shall be 1/4 inch in size. Letters shall be all upper case. Label paper shall be white paper having a smooth finish. Ink shall be commercial waterproof and black. Exterior packing boxes shall be marked in accordance with Dwg. 8796522. The ICC nomenclature shall be "Small Arms Ammunition."

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection Equipment Designs.-Inspection equipment designs are of two types-Government Special Inspection Equipment



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(SIE) designs and Contractor Designs. SIE designs, if available are designated by drawing numbers (on the Equipment Lists referenced on the Equipment Tabulation) or (under the "Method of Inspection" headings in Section 4). Design responsibility for all other inspection equipment is assigned to the Contractor. Equipment to be designed by the Contractor is designated "CD". The Contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the Contractor may submit alternate or modified Contractor designs of SIE in accordance with 6.2.2 and 6.2.3, should he elect to do so.

6.2.1 SIE Designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the Contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor Designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified, and may include commercial equipment which the Contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs, if not otherwise covered thereby (See 6.2.1 c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The Contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each Contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.



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6.2.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.3 New Material Test Sample Requirements.-All requests for permission to use materials other than those specified on applicable drawings or specifications should be accompanied by a sample of the proposed substitute material to permit evaluation by Picatinny Arsenal. Prior to approval, all substitute materials must be tested for compatibility with M9 Propellant in addition to such testing required to evaluate the performance of the proposed substitute.

6.4 Submission of test data.-In addition to the normal distribution of records, when the ignition cartridge is procured by the Department of the Army, one copy of all ballistic test data shall be forwarded to the following:

Commander
Picatinny Arsenal
ATTN: SMUPA-QA-A-A
Dover, New Jersey 07801

6.5 Check test for possible deterioration of Cartridge, Ignition.-The following test is designed to prevent the loading of major ammunition items with cartridges which have deteriorated. The test shall be performed on each lot of cartridges which has been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the cartridge is assembled does not require that the check test be performed, the contracting officer shall insert a provision for the performance of the check test in the contract. The contracting officer shall arrange to have the contractor reimbursed for any expense incurred in the performance of the check test for possible deterioration.

6.5.1 Velocity and functioning.-A sample shall be selected in accordance with MIL-STD-414, Section B Table B-3, Code Letter M and an AQL 1.5 percent. The test shall be performed as specified in 4.4.4.

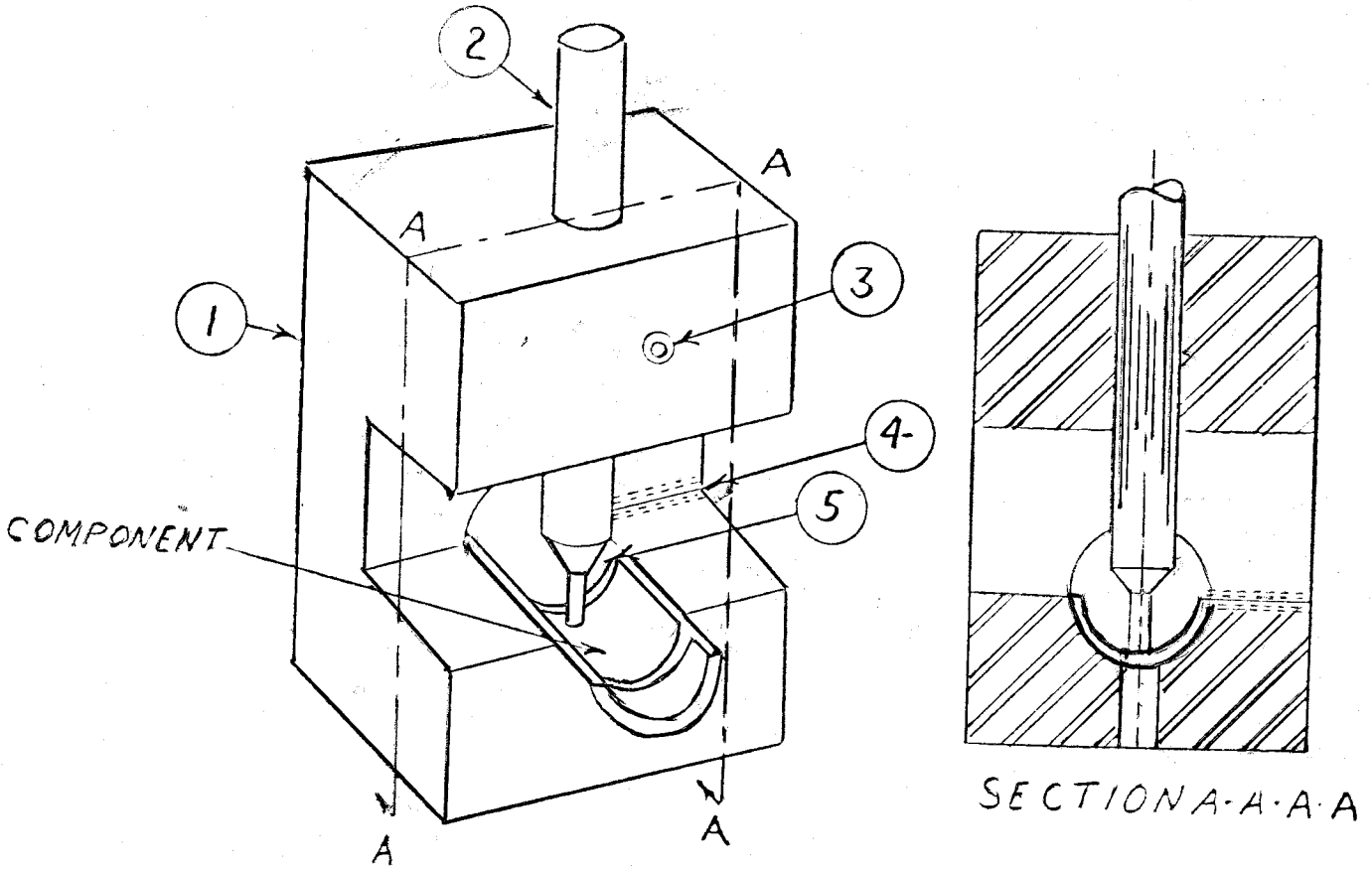
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ARMY-MU

Project Number: 1315-A-026



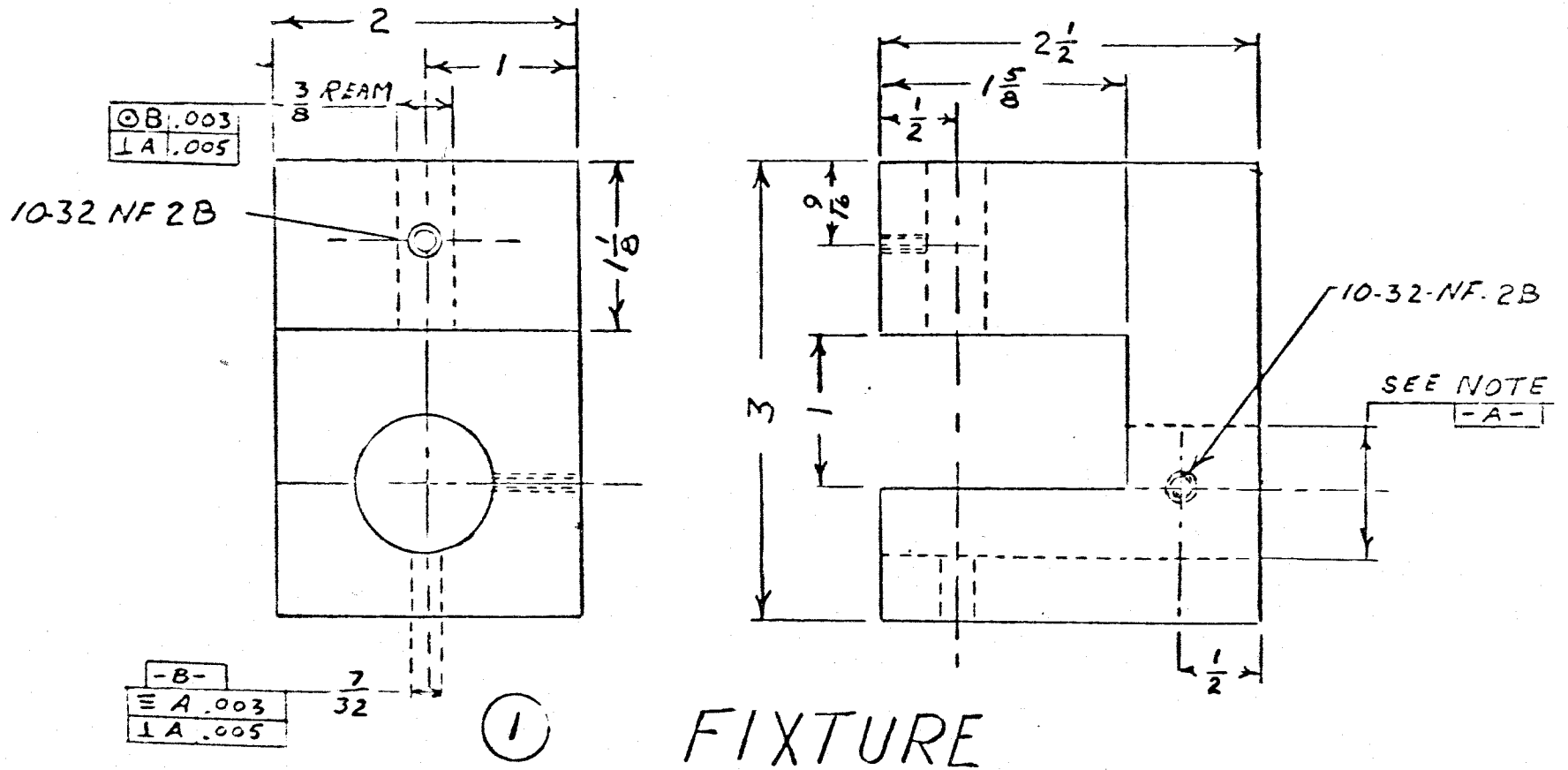
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ASSEMBLY
BURSTING STRENGTH
TEST FIXTURE

FIGURE 1 SHEET 1

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(1) FIXTURE

MAT'L - MACH. STEEL 1 REQ'D

(3) 10-32 SOC. SETSCREW FULL DOG PT. 3/8 LG.

(4) 10-32 NF-2A SOC. SETSCREW FULL DOG PT. 1/2 LG

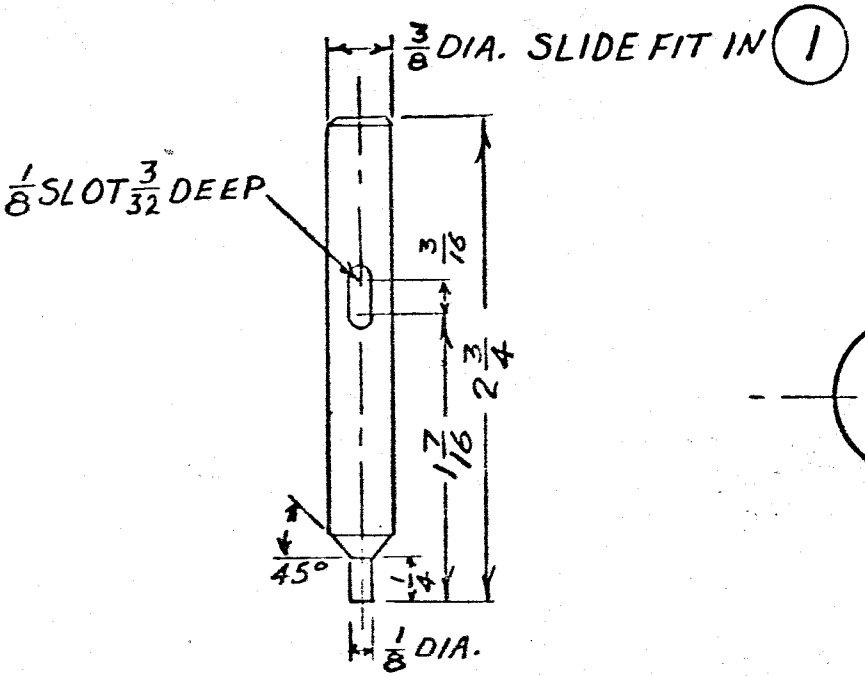
NOTE: DIAMETER TO SUIT APPLICABLE BODY ASS'Y

DIMENSIONS ARE IN INCHES

TOLERANCES ON – DECIMALS ± .005
FRACTIONS ± 1/64
ANGLES ± 2°

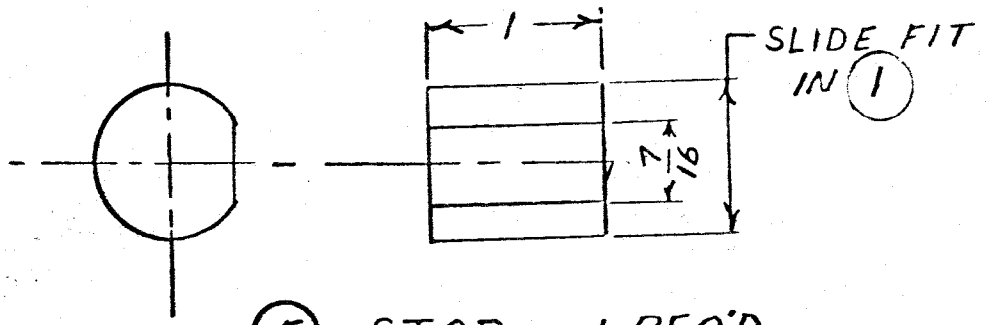
FIGURE 1 SHEET 2

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(2) PUNCH

MATERIAL, DRILL ROD
1 REQ'D.



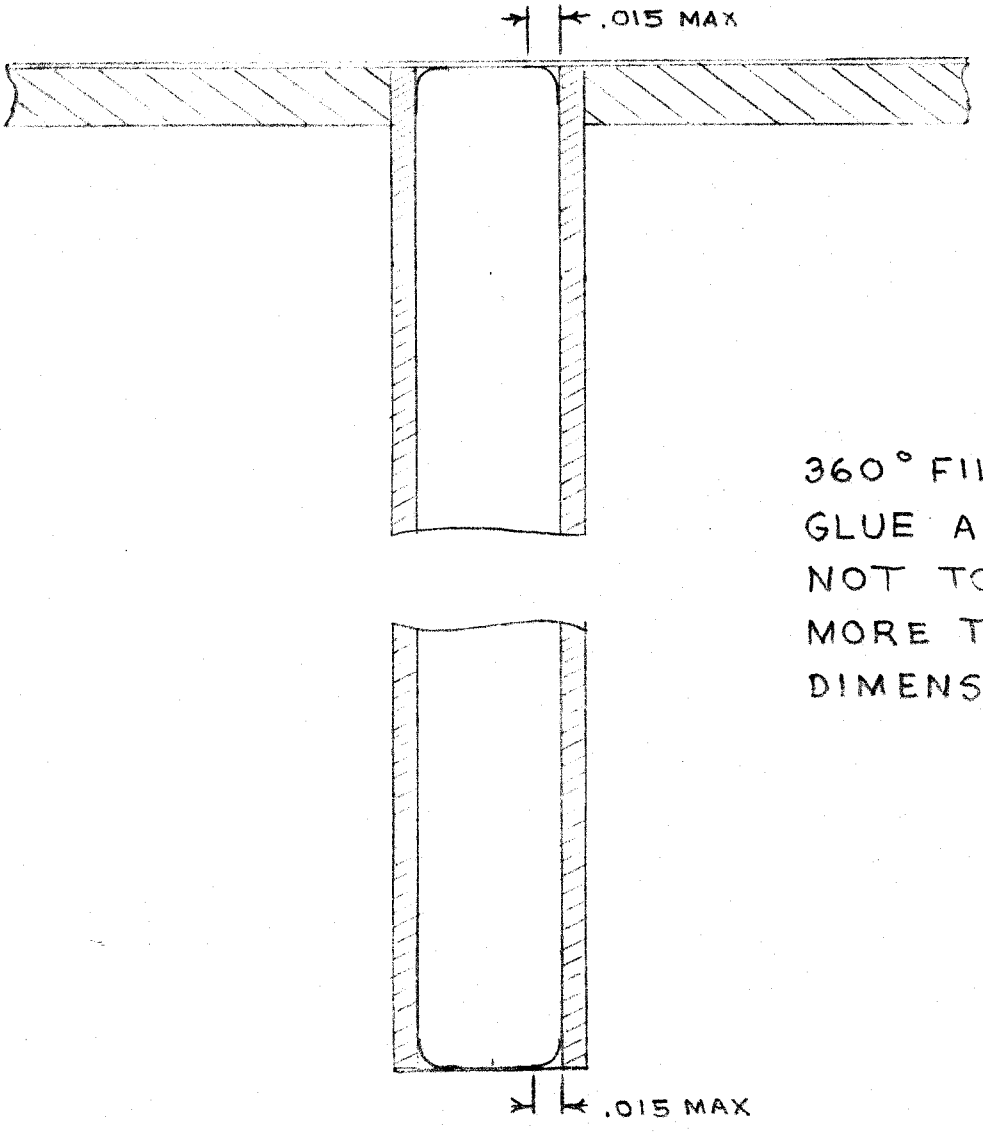

(5) STOP 1 REQ'D

MAT'L - MACH. STEEL



FIGURE 1 SHEET 8

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TUBE ASSEMBLY

FIGURE 2

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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

**DD FORM 1 JAN 66 1426** REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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